

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416400
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Pu; Yingbin et al.

Lighting effect module and light

Abstract

A lighting effect module includes a light module and a connection assembly. The light module includes a housing, a light cover, a female connector, a light-emitting body, and a circuit board. The light cover and the female connector are each disposed on the housing. The light-emitting body and the circuit board are each disposed in the housing. The light-emitting body and the female connector are electrically connected to the circuit board. The connection assembly includes an electrical connection module and an engaging module. The electrical connection module is configured to be plugged in and cooperate with female connectors of two light modules. The engaging module is configured to engage with housings of the two light modules to enable the two light modules to be electrically connected and engaged through the connection assembly.

Inventors:	Pu; Yingbin (Shenzhen, CN), Yang; Xiangpiao (Shenzhen, CN)
Applicant:	Luxshare Precision Industry Company Limited (Shenzhen, CN)
Family ID:	1000008037708
Assignee:	Luxshare Precision Industry Company Limited (Shenzhen, CN)
Appl. No.:	18/768638
Filed:	July 10, 2024

Foreign Application Priority Data

CN	202410470610.1	Apr. 17, 2024
----	----------------	---------------

Publication Classification

Int. Cl.: F21V23/06 (20060101); F21V21/002 (20060101); F21V23/00 (20150101)

U.S. Cl.:

Field of Classification Search

CPC: F21V (23/06); F21V (21/002); F21V (23/005)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5797767	12/1997	Schell	439/490	H01R 12/79
7494244	12/2008	Van Diep	439/502	F21V 19/001
9528661	12/2015	Ferro	N/A	F21K 9/90
2006/0262533	12/2005	Lin	362/294	F21V 29/70
2009/0196034	12/2008	Gherardini	362/235	F21V 21/005
2015/0129014	12/2014	Shu	136/251	F24S 25/13
2018/0202636	12/2017	Fox	N/A	F21V 23/002

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
211289746	12/2019	CN	N/A

Primary Examiner: Dzierzynski; Evan P

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims the priority to Chinese Patent Application No. CN202410470610.1, filed on Apr. 17, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present application relates to the technical field of lights, in particular, a lighting effect module and a light.

BACKGROUND

(3) Electronic devices such as business machines, servers, and audio are all equipped with light groups. The light groups may be indicator lights and ambient lights. The light groups may be installed on the front panel, side panel, and inside of the electronic devices. The installation position and installation quantity may be set according to the actual requirements of consumers.

(4) In the related art, light groups include multiple light modules connected in series. To improve the aesthetics of light groups, light modules are connected by magnetic attraction, and two adjacent light modules are connected by a connector. Since a magnet needs to be disposed in each light module, the overall size of the light module is relatively large and takes up a larger space. Light modules may also be connected to each other by the plugging of male connectors and female connectors. However, the connection between light modules completely depends on the fixedness of the male connectors, and since the male connectors bear the force caused by the up, down, left, and right shaking between modules for a long time, the connection between the male connectors and the female connectors fails. Thus, signal transmission is affected, and reliability is lowered.

(5) Therefore, a light module and a light are urgently needed to resolve the preceding problem.

SUMMARY

(6) One object of the present application is to provide a lighting effect module. The lighting effect module, with relatively high connection reliability, reduces the probability of connection failure and ensures the stability of signal transmission.

(7) As conceived above, the present application provides the solutions below.

(8) A lighting effect module includes a light module and a connection assembly.

(9) The light module includes a housing, a light cover, a female connector, a light-emitting body, and a circuit board. The light cover and the female connector are each disposed on the housing. The light-emitting body and the circuit board are each disposed in the housing. The light-emitting body and the female connector are electrically connected to the circuit board.

(10) The connection assembly includes an electrical connection module and an engaging module. The electrical connection module is configured to be plugged in and cooperate with female connectors of two light modules. The engaging module is configured to engage with the housings of the two light modules to enable the two light modules to be electrically connected and engaged through the connection assembly.

(11) Optionally, the engaging module includes a connection block and multiple first engaging pieces disposed on two sides of the connection block, the housing is provided with an engaging hole, and a first engaging piece of the multiple first engaging pieces disposed at one side of the connection block is configured to engage and cooperate with the engaging hole.

(12) Optionally, the multiple first engaging pieces are disposed on the opposite sides of the connection block.

(13) Optionally, a shallow groove is disposed on the housing and is configured to limit the connection block.

(14) Optionally, the electrical connection module includes two male connectors that are disposed on the engaging module and electrically connected to each other, and a male connector is configured to be plugged in and cooperate with the female connector.

(15) Optionally, the female connector includes a seat body and a conductive terminal.

(16) The seat body is formed with a slot, a first plug hole, and an auxiliary slot. The first plug hole separately communicates with the slot and the auxiliary slot.

(17) The conductive terminal includes a terminal body and a limiting structure. The terminal body is inserted through the first plug hole. One end of the terminal body has a contact portion disposed in the slot. The electrical connection module is configured to be plugged into the slot and electrically connected to the contact portion. Another end of the terminal body is connected to the limiting structure limited in the auxiliary slot. The conductive terminal is electrically connected to the circuit board.

(18) Optionally, the female connector also includes an engaging holder, the seat body is further formed with a second plug hole, and the electrical connection module is formed with a clamping groove, where one end of the engaging holder is configured to be limited in the second plug hole and another end of the engaging holder is configured to engage and cooperate with the clamping groove.

(19) Optionally, a second engaging piece is disposed on the female connector and is configured to engage and cooperate with the circuit board.

(20) Optionally, the cross section of the housing is quadrilateral, and each of at least one sidewall of the housing is provided with the female connector.

(21) Another object of the present application is to provide a light. The light, with relatively high flexibility and connection reliability, reduces the probability of connection failure.

(22) As conceived above, the present application provides the solutions below.

(23) A light includes the preceding lighting effect module.

(24) Multiple light modules and multiple connection assemblies are provided. Two light modules

are electrically connected and engaged through a connection assembly.

(25) The beneficial effects of the present application are described below.

(26) In the lighting effect module and the light provided by the present application, a female connector is disposed on the housing of a light module, and an electrical connection module of the connection assembly is configured to be plugged in and cooperate with the female connector to implement the electrical connection between two light modules that need to be spliced; the connection assembly also includes an engaging module configured to engage with the housings of two light modules so that the two light modules can also be engaged through the connection assembly, and magnets are not required to be installed in the light modules, thereby making the overall size of the light modules smaller. The configuration of the engaging module enables the connection between light modules to be achieved simultaneously by the electrical connection module and the engaging module, reduces the force borne by the electrical connection module, and thus reduces the probability of failure of the electrical connection module and the female connector being plugged. Moreover, the stability of signal transmission is ensured, enabling the lighting effect module to have higher reliability.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a view one illustrating the structure of a light module according to an embodiment of the present application.

(2) FIG. 2 is an exploded view one illustrating the structure of a light module according to an embodiment of the present application.

(3) FIG. 3 is a sectional view one of a light module according to an embodiment of the present application.

(4) FIG. 4 is a view two illustrating the structure of a light module according to an embodiment of the present application.

(5) FIG. 5 is an exploded view two illustrating the structure of a light module according to an embodiment of the present application.

(6) FIG. 6 is a sectional view two of a light module according to an embodiment of the present application.

(7) FIG. 7 is a sectional view three of a light module according to an embodiment of the present application.

(8) FIG. 8 is a view one illustrating the structure of a female connector according to an embodiment of the present application.

(9) FIG. 9 is a view two illustrating the structure of a female connector according to an embodiment of the present application.

(10) FIG. 10 is a view one illustrating the structure of a seat body according to an embodiment of the present application.

(11) FIG. 11 is a view two illustrating the structure of a seat body according to an embodiment of the present application.

(12) FIG. 12 is a view illustrating the structure of a conductive terminal according to an embodiment of the present application.

(13) FIG. 13 is a view illustrating the structure of an engaging holder according to an embodiment of the present application.

REFERENCE LIST

(14) **100** light module **110** housing **111** engaging hole **1111** small hole section **1112** large hole section **112** shallow groove **113** through hole **120** light cover **130** female connector **131** seat body **1311** slot **1312** first plug hole **1313** auxiliary slot **1314** second plug hole **1315** receiving slot **132**

conductive terminal **1321** terminal body **13211** contact portion **13212** buffer groove **1322** limiting structure **133** engaging holder **1331** engaging protrusion **140** light-emitting body **150** circuit board **160** second engaging piece **200** connection assembly **210** electrical connection module **211** male connector **212** clamping groove **220** engaging module **221** connection block **222** first engaging piece **2221** block X first direction Z second direction Y third direction
DETAILED DESCRIPTION

(15) To make problems to be solved, adopted solutions, and achieved effects of the present application clearer, solutions of the present application are further described below through embodiments in conjunction with drawings. It is to be understood that the embodiments described herein are intended to explain the present application and not to limit the present application. In addition, it should be noted that for ease of description, only the part, instead of all, related to the present application is illustrated in the drawings.

(16) It is to be noted that similar reference numerals and letters indicate similar items in subsequent drawings. Therefore, once a certain item is defined in one drawing, the item needs no more definition and explanation in the subsequent drawings.

(17) In the description of the present application, unless otherwise expressly specified and limited, the term “connected to each other”, “connected”, or “secured” is to be construed in a broad sense, for example, as securely connected, detachably connected, or integrated; mechanically connected or electrically connected; directly connected to each other or indirectly connected to each other via an intermediary; or internally connected between two components or interaction relations between two components. For those of ordinary skill in the art, specific meanings of the preceding terms in the present application may be construed based on specific situations.

(18) In the present application, unless otherwise expressly specified and limited, when a first feature is described as “above” or “below” a second feature, the first feature and the second feature may be in direct contact or be in contact via another feature between the two features. Moreover, when the first feature is “on”, “above”, or “over” the second feature, the first feature is right on, above, or over the second feature, or the first feature is obliquely on, above, or over the second feature, or the first feature is simply at a higher level than the second feature. When the first feature is “under”, “below”, or “underneath” the second feature, the first feature is right under, below, or underneath the second feature, or the first feature is obliquely under, below, or underneath the second feature, or the first feature is simply at a lower level than the second feature. In the description of the embodiments, unless otherwise specified, “multiple” mentioned herein means two or more.

(19) In the description of the embodiments, the orientation or position relationships indicated by terms “above”, “below”, “right”, and the like are based on the orientation or position relationships shown in the drawings, only for ease of description and simplifying an operation, and these relationships do not indicate or imply that the referred device or element has a specific orientation and is constructed and operated in a specific orientation, and thus it is not to be construed as limiting the present application. In addition, terms “first” and “second” are used only to distinguish between descriptions and have no special meaning.

(20) The embodiment provides a lighting effect module, with relatively high connection reliability, reduces the probability of connection failure and ensures the stability of signal transmission.

(21) As shown in FIG. 1, a lighting effect module includes a light module **100** and a connection assembly **200**, and the connection assembly **200** is electrically connected to and engaged with the light module **100** to implement the electrical connection and engaging between two light modules **100**.

(22) As shown in FIG. 2 and FIG. 3, the light module **100** includes a housing **110**, a light cover **120** disposed on the housing **110**, a female connector **130** disposed on the housing **110**, and a light-emitting body **140** and a circuit board **150** that are both disposed in the housing **110**. The female connector **130** is electrically connected to the circuit board **150** to transmit a signal to the circuit

board **150**. The light-emitting body **140** is electrically connected to the circuit board **150** so that the light-emitting body **140** can be controlled to emit light or not emit light by the circuit board **150**.

(23) The connection assembly **200** includes an electrical connection module **210** and an engaging module **220**. The electrical connection module **210** is configured to be plugged in and cooperate with the female connectors **130** of two light modules **100** to implement the electrical connection between the two light modules **100**. The engaging module **220** is configured to engage with the housings **110** of two light modules **100** so that the housings **110** of the two light modules **100** can be engaged through the engaging module **220**. The arrangement of the connection assembly **200** achieves the electrical connection and engaging between light modules **100**. The connection assembly **200** has rich functions.

(24) For the lighting effect module provided by the embodiment, a female connector **130** is disposed on the housing **110** of a light module **100**, and an electrical connection module **210** of the connection assembly **200** is configured to be plugged in and cooperate with the female connector **130** to implement the electrical connection between two light modules **100** that need to be spliced; the connection assembly **200** also includes an engaging module **220** configured to engage with the housings **110** of two light modules **100** so that the two light modules **100** can also be engaged through the connection assembly **200**, and magnets are not required to be installed in the light modules **100**, thereby making the overall size of the light modules **100** smaller. The configuration of the engaging module **220** enables the connection between light modules **100** to be achieved simultaneously by the electrical connection module **210** and the engaging module **220**, reduces the force borne by the electrical connection module **210**, and thus reduces the probability of failure of the electrical connection module **210** and the female connector **130** being plugged. Moreover, the stability of signal transmission is ensured, enabling the lighting effect module to have higher reliability.

(25) Optionally, as shown in FIG. 2, the engaging module **220** includes a connection block **221** and a first engaging piece **222**. At least two first engaging pieces **222** are provided, and first engaging pieces **222** are disposed on two sides of the connection block **221**. Preferably, the first engaging pieces **222** are disposed on the opposite sides of the connection block **221**. Certainly, it can be understood that the first engaging pieces **222** are disposed on two adjacent sides of the connection block **221**. A first engaging piece **222** on one side of the connection block **221** is configured to engage with the housing **110** of one light module **100**, and a first engaging piece **222** on another side of the connection block **221** is configured to engage with the housing **110** of another light module **100**. Illustratively, an engaging hole **111** is disposed on the housing **110**, and the first engaging piece **222** can be inserted into the engaging hole **111** and engage and cooperate with the engaging hole **111**. Optionally, the engaging hole **111** is disposed on the sidewall of the housing **110** to facilitate the insertion of the first engaging piece **222**. Certainly, it can be understood that the engaging hole **111** may also be disposed on the bottom wall of the housing **110**, which is not limited by this embodiment.

(26) In some optional embodiments, as shown in FIG. 5, two blocks **2221** arranged opposite to each other are disposed at an end of the first engaging piece **222**, and a gap exists between the two blocks **2221** so that the two blocks **2221** can elastically deform. That is, the two blocks **2221** can be close to each other under an external force, and when the external force disappears, the two blocks **2221** can return to the original state.

(27) As shown in FIG. 4 and FIG. 5, the engaging hole **111** in this embodiment includes a small hole section **1111** and a large hole section **1112**. In the case where the engaging hole **111** is disposed in the sidewall of the housing **110**, the small hole section **1111** is closer to the sidewall of the housing **110** than the large hole section **1112**. The two blocks **2221** need to be deformed when passing through the small hole section **1111**, that is, the two blocks **2221** are close to each other under the extrusion of the hole walls of the small hole section **1111**. When the two blocks **2221** move to the large hole section **1112**, the two blocks **2221** return to their original shape and can be

limited in the large hole section **1112**, thereby achieving the engaging between the first engaging piece **222** and the housing **110**. It should be noted that when the two blocks **2221** are located in the large hole section **1112**, and when the first engaging piece **222** is pulled in the axis direction of the engaging hole **111**, the end surfaces of the blocks **2221** abut against the housing **110** so that the two blocks **2221** do not enter the small hole section **1111**. In this manner, the engaging reliability of the first engaging piece **222** and the housing **110** is ensured.

(28) Optionally, as shown in FIG. 4, the large hole section **1112** extends to the bottom wall of the housing **110** so that a user may use a tool to hold two blocks **2221** close to each other through the large hole section **1112**. In this case, when the first engaging piece **222** is pulled outward, the two blocks **2221** can smoothly enter the small hole section **1111**, thereby implementing the disassembly of the first engaging piece **222** and the housing **110** and thus facilitating the reassembly of the lighting effect module and the reuse of the connection assembly **200**. Thus, relatively high flexibility is provided.

(29) In the embodiment, multiple first engaging pieces **222** are disposed on the opposite sides of the connection block **221** to improve the engaging effect and engaging stability between the engaging module **220** and the housing **110**. In the embodiment, as shown in FIG. 2, two first engaging pieces **222** are disposed on each side of the connection block **221**, and the two first engaging pieces **222** are disposed at intervals in the length direction of the connection block **221**.

(30) In the embodiment, with reference to FIG. 2, a shallow groove **112** is disposed in the housing **110**. Specifically, a shallow groove **112** is disposed on the surface of the housing **110** on which the engaging hole **111** is disposed, and the shallow groove **112** is used to limit the connection block **221**, thereby further reducing the probability of the connection assembly **200** being shaken and improving the connection reliability between the light module **100** and the connection assembly **200**. It should be noted that the shallow groove **112** limits the connection block **221** in the length and width directions of the connection block **221** but does not limit the connection block **221** in the thickness direction of the connection block **221** to facilitate the connection and detachment of the connection assembly **200** and the light module **100**. Preferably, the depth of the shallow groove **112** is less than the thickness of connection block **221** to avoid the interference between the housings **110** of the two light modules **100**.

(31) Optionally, as shown in FIG. 2, the electrical connection module **210** includes two male connectors **211** that are disposed on the engaging module **220** and electrically connected to each other, so as to implement power transmission. It should be noted that the two male connectors are disposed on the two sides of the connection block **221** on which the first engaging pieces **222** are disposed. That is, one side of the connection block **221** has one male connector **211** and two first engaging pieces **222**, and another side of the connection block **221** also has a male connector **211** and two first engaging pieces **222**. In the embodiment, the male connector **211** is disposed between two first engaging pieces **222**. The male connector **211** can be plugged in and cooperate with the female connector **130** to implement the electrical connection between the male connector **211** and the female connector **130**. In the embodiment, the male connector **211** may include a male terminal body and a conductor disposed on the male terminal body, and the conductors of the two male connectors **211** are electrically connected.

(32) In some optional embodiments, the cross section of the housing **110** is quadrilateral so that the volume of the housing **110** can be smaller. In addition, at least one side of the housing **110** is provided with a female connector **130**. In the embodiment, all four sidewalls of the housing **110** are provided with a female connector **130** to improve the splicing flexibility of the light module **100**. It can be understood that the cross section of the housing **110** may also be polygonal, circular, and elliptical, which is not limited in the embodiment. When the cross section of the housing **110** is circular or oval, two sides of the housing **110** are one part and another part of the side surface of the housing **110**.

(33) Optionally, as shown in FIG. 6 to FIG. 13, the female connector **130** includes a seat body **131**

and a conductive terminal **132**. A slot **1311**, a first plug hole **1312**, and an auxiliary slot **1313** are opened on the seat body **131**. The first plug hole **1312** separately communicates with the slot **1311** and the auxiliary slot **1313**. In the embodiment, a through hole **113** is disposed on the sidewall of the housing **110**, the through hole **113** communicates with the slot **1311** of the female connector **130**, and the male connector **211** passes through the through hole **113** and is inserted into the slot **1311**.

(34) The conductive terminal **132** includes a terminal body **1321** and a limiting structure **1322**. The terminal body **1321** is inserted through the first plug hole **1312**, and one end of the terminal body **1321** has a contact portion **13211** disposed in the slot **1311**. The electrical connection module **210** is configured to be inserted into the slot **1311** and electrically connected to the contact portion **13211**. The limiting structure **1322** is connected to another end of the terminal body **1321** and is limited in the auxiliary slot **1313**. The conductive terminal **132** is electrically connected to the circuit board **150**.

(35) As shown in FIG. 6 to FIG. 13, the female connector **130** includes a seat body **131** and a conductive terminal **132**. A slot **1311**, a first plug hole **1312**, and an auxiliary slot **1313** are opened on the seat body **131**. Specifically, the slot **1311** is opened on an end surface of one end of the seat body **131** and is configured to be inserted by a male connector **211**. The first plug hole **1312** extends from an end surface of another end of the seat body **131** in the length direction of the seat body **131**, and the first plug hole **1312** separately communicates with the slot **1311** and the auxiliary slot **1313**. The auxiliary slot **1313** is spaced apart from the slot **1311**, that is, the two are not directly connected.

(36) The conductive terminal **132** includes a terminal body **1321** and a limiting structure **1322**. The terminal body **1321** is inserted through the first plug hole **1312**, and one end of the terminal body **1321** has a contact portion **13211** disposed in the slot **1311**. The contact portion **13211** is configured to be electrically contacted to the male connector **211**. The limiting structure **1322** is connected to another end of the terminal body **1321** and is limited in the auxiliary slot **1313**, that is, the limiting structure **1322** and the terminal body **1321** form a U-shaped structure so that the conductive terminal **132** can be sandwiched in the seat body **131** to achieve the fixation of the conductive terminal **132** in the seat body **131**.

(37) In the female connector **130** provided in the embodiment, an auxiliary slot **1313** is disposed on the seat body **131**; the conductive terminal **132** includes a terminal body **1321** and a limiting structure **1322**; the terminal body **1321** is inserted through the first plug hole **1312**, and one end of the terminal body **1321** has a contact portion **13211** disposed in the slot **1311**; the limiting structure **1322** is connected to another end of the terminal body **1321** and is limited in the auxiliary slot **1313**. In this manner, the limitation of the terminal body **1321** in the seat body **131** is indirectly achieved so that when the male connector **211** is inserted into the slot **1311**, the terminal body **1321** does not move in the plugging and unplugging direction. Thus, it is ensured that the contact portion **13211** of the terminal body **1321** can be smoothly connected to the male connector **211** so that the assembly and fixation of the conductive terminal **132** on the seat body **131** does not require the contact with an inner insulation block, and the conductive terminal **132** is fixed in the seat body **131** through its own structure. Thus, the overall size of the female connector **130** can be smaller, thereby making the overall size of the light module **100** smaller. Then it is convenient for the light module **100** to be applied in electronic devices with high space requirements.

(38) Optionally, as shown in FIG. 8 and FIG. 9, the female connector **130** also includes an engaging holder **133**. As shown in FIG. 11, a second plug hole **1314** is also opened on the seat body **131**; one end of the engaging holder **133** is limited in the second plug hole **1314** so that the engaging holder **133** does not move in the plugging and unplugging direction; another end of the engaging holder **133** is configured to engage and cooperate with the male connector **211** to improve the stability and reliability of the contact between the female connector **130** and the male connector **211**.

(39) Illustratively, as shown in FIG. 12, an engaging protrusion **1331** is disposed on another end of the engaging holder **133**, a clamping groove **212** is disposed on the electrical connection module **210**, and the another end of the engaging holder **133** engages and cooperate with the clamping groove **212**. Specifically, as shown in FIG. 4, a clamping groove **212** is disposed on the male connector **211**, and the engaging protrusion **1331** engages and cooperates with the clamping groove **212** so that the male connector **211** can be limited in the plugging and unplugging direction, and thus the male connector **211** can be in stable contact with the contact portion **13211**.

(40) In some optional embodiments, the engaging holder **133** is a ground terminal. That is, when an electronic product has a high current, the engaging holder **133** may be used as a power ground terminal so that the female connector **130** can be used to transmit large currents, which further improves the applicable scope of the female connector **130** and the light module **100** using the female connector **130**. Illustratively, the engaging holder **133** is made of copper alloy to have greater signal transmission capability.

(41) Optionally, two engaging holders **133** and two second plug holes **1314** are provided in a one-to-one correspondence, the two engaging holders **133** are disposed opposite to each other in a first direction X, and one end of each engaging holder **133** is limited in the second plug hole **1314** corresponding to the each engaging holder **133**. By the configuration of two engaging holders **133**, the male connector **211** can be limited and engaged in the first direction X, thereby further improving the stability of the connection between the male connector **211** and the female connector **130**. Illustratively, one of the two engaging holders **133** is disposed on one side of the slot **1311** in the first direction X, and the other engaging holder is disposed on another side of the slot **1311** in the first direction X.

(42) Further, optionally, receiving slots **1315** are opened on the opposite sides of the seat body **131** in the first direction X. The receiving slot **1315** extends in the first direction X to communicate with the slot **1311**. The two engaging holders **133** are in a one-to-one correspondence with the two receiving slots **1315**. The another end of each engaging holder **133** can enter the receiving slot **1315** corresponding to the each engaging holder **133** under the squeeze of the male connector **211** to form a temporary avoidance, thereby facilitating the insertion of the male connector **211**. In the embodiment, one end of the engaging holder **133** is fixed, and another end of the engaging holder **133** is movable in the first direction X, that is, the engaging holder **133** can elastically deform to facilitate the engaging and cooperation between the engaging protrusion **1331** and the clamping groove **212**.

(43) Optionally, a first barb structure is disposed on a portion of the engaging holder **133** disposed in the second plug hole **1314**, and the first barb structure abuts against the hole wall of the second plug hole **1314** to prevent the engaging holder **133** from moving in the plugging and unplugging direction during the plugging and unplugging process of the male connector **211**. Thus, the strength of the connection between the engaging holder **133** and the seat body **131** is increased, and the probability of the engaging holder **133** becoming loose is reduced. In the embodiment, the first barb structure is a barb-shaped protrusion bent toward another end of the seat body **131**.

Illustratively, the first barb structure is a ring protrusion.

(44) In this embodiment, a second barb structure is disposed on the limiting structure **1322**, and the second barb structure abuts against the slot wall of the auxiliary slot **1313** to prevent the limiting structure **1322** from moving in the plugging and unplugging direction during the plugging and unplugging process of the male connector **211**. Thus, the strength of the connection between the limiting structure **1322** and the seat body **131** is increased, and the probability of the limiting structure **1322** becoming loose is reduced. In the embodiment, the second barb structure is a barb-shaped protrusion bent toward another end of the seat body **131**. Illustratively, the second barb structure is disposed on a side of the limiting structure **1322** away from the terminal body **1321**.

(45) Optionally, as shown in FIG. 12, a buffer slot **13212** is disposed on one end of the terminal body **1321** connected to the limiting structure **1322**. The configuration of the buffer slot **13212**

enables the end of the terminal body **1321** disposed with the contact portion **13211** to move in a second direction Z so that the terminal body **1321** can be in elastic contact with the male connector **211**, thereby improving the adaptation capability and contact stability of the conductive terminal **132**. The second direction Z is perpendicular to the first direction X. In the embodiment, the first direction X is the width direction of the seat body **131**, and the second direction Z is the height direction of the seat body **131**.

(46) Optionally, as shown in FIG. 9, multiple conductive terminals **132** and one slot **1311** are provided, multiple first plug holes **1312** and auxiliary slots **1313** are provided in a one-to-one correspondence, and contact portions **13211** at one end of multiple terminal bodies **1321** are all disposed in the slot **1311** to be in electrical contact with the male connector **211** disposed in the slot **1311**. Multiple terminal bodies **1321** are in a one-to-one correspondence with multiple first plug holes **1312**, and each terminal body **1321** is inserted through a corresponding first plug hole **1312** so that the multiple terminal bodies **1321** can be independent of each other and do not contact each other. Multiple limiting structures **1322** are in a one-to-one correspondence with multiple auxiliary slots **1313**, and each limiting structure **1322** is limited in a corresponding auxiliary slot **1313** so that the auxiliary slot **1313** can limit the limiting structure **1322** in each direction, thereby improving the limiting effect of each limiting structure **1322**.

(47) Illustratively, the conductive terminal **132** has an integrated structure, that is, the terminal body **1321** and the limiting structure **1322** are integrally formed to have relatively high structural strength. By the configuration of multiple auxiliary slots **1313**, the electrical contact between limiting structures **1322** can be avoided, and thus the electrical contact between conductive terminals **132** can be avoided.

(48) Further, optionally, multiple conductive terminals **132** are spaced apart in the first direction X, and the auxiliary slot **1313** and a corresponding first plug hole **1312** are spaced apart in the second direction Z so that the configuration of the auxiliary slot **1313** does not affect the spacing between the conductive terminals **132** and allows the size of the female connector **130** in the first direction X to be smaller.

(49) In the embodiment, the auxiliary slot **1313** is disposed on a side of the slot **1311** in a third direction Y, and the first plug hole **1312** is disposed under the auxiliary slot **1313** so that the space on the seat body **131** can be fully utilized, and the miniaturization of the female connector **130** is facilitated. It should be noted that any two of the first direction X, the second direction Z, and the third direction Y are perpendicular to each other. The third direction Y may be the length direction of the seat body **131**.

(50) Optionally, the spacing between two adjacent conductive terminals **132** in the first direction X is 0.8 mm to 2 mm. Illustratively, the spacing between two adjacent conductive terminals **132** in the first direction X is 0.8 mm, 1 mm, 1.5 mm, or 2 mm. Preferably, the spacing between two adjacent conductive terminals **132** in the first direction X is 1 mm.

(51) Optionally, the length of the seat body **131** is 5 cm to 10 cm. Illustratively, the length of the seat body **131** is 5 cm, 6 cm, or 10 cm. Preferably, the length of the seat body **131** is 6.55 cm. It should be noted that the length direction of the seat body **131** is the third direction Y.

(52) The width of the seat body **131** is 5.5 cm to 10 cm. Illustratively, the width of the seat body **131** is 5 cm, 6 cm, or 10 cm. Preferably, the width of the seat body **131** is 6 cm. It should be noted that the width direction of the seat body **131** is the first direction X.

(53) The height of the seat body **131** is 2 cm to 5 cm. Illustratively, the height of the seat body **131** is 5 cm, 6 cm, or 10 cm. Preferably, the height of the seat body **131** is 2.9 cm. It should be noted that the height direction of the seat body **131** is the second direction Z.

(54) In the embodiment, the height of the slot **1311** is 1 cm to 2 cm. Preferably, the height of the slot **1311** is 1.3 cm. The height of the slot **1311** is 4 cm to 6 cm. Preferably, the height of the slot **1311** is 4.5 cm.

(55) In some embodiments, as shown in FIG. 8, a second engaging piece **160** is disposed at the

bottom of the seat body **131**, and the second engaging piece **160** engages and cooperates with the circuit board **150** so that the female connector **130** can better contact the contact point on the circuit board **150**. Illustratively, the circuit board **150** is provided with a hole, and the second engaging piece **160** passes through the hole and engages with the circuit board **150**. In this embodiment, the specific structure of the second engaging piece **160** is the same as that of the first engaging piece **222**. When entering the hole, two blocks **2221** of the second engaging piece **160** are squeezed by the hole wall and then approach each other.

(56) In this embodiment, a contact point is disposed on the circuit board **150**, the conductive terminal **132** has a first connection end disposed outside the seat body **131**, and the first connection end is electrically connected to the contact point on the circuit board **150**. The engaging holder **133** has a second connection end disposed outside the seat body **131**. The second connection end is electrically connected to another contact point on the circuit board **150**.

(57) The embodiment also provides a light, including the preceding light module.

(58) Multiple light modules **100** and connection assemblies **200** are provided. Two light modules **100** are electrically connected and engaged by a connection assembly **200**. The light provided by the embodiment has relatively high flexibility and connection reliability and reduces the probability of connection failure.

(59) It is to be noted that the preceding are only preferred embodiments of the present application and technical principles used therein. It is to be understood by those skilled in the art that the present application is not limited to the embodiments described herein. Those skilled in the art can make various apparent modifications, adaptations, and substitutions without departing from the scope of the present application. Therefore, while the present application is described in detail through the preceding embodiments, the present application is not limited to the preceding embodiments and may include more other equivalent embodiments without departing from the concept of the present application. The scope of the present application is determined by the scope of the appended claims.

Claims

1. A lighting effect module, comprising: a light module, wherein the light module comprises a housing, a light cover, a female connector, a light-emitting body, and a circuit board, the light cover and the female connector are each disposed on the housing, the light-emitting body and the circuit board are each disposed in the housing, and the light-emitting body and the female connector are electrically connected to the circuit board; and a connection assembly, wherein the connection assembly comprises an electrical connection module and an engaging module, the electrical connection module is configured to be plugged into and cooperate with female connectors of two light modules, and the engaging module is configured to engage with housings of the two light modules to enable the two light modules to be electrically connected and engaged through the connection assembly; wherein the female connector comprises: a seat body formed with a slot, a first plug hole, and an auxiliary slot, and the first plug hole separately communicates with the slot and the auxiliary slot and a conductive terminal comprising a terminal body and a limiting structure, wherein the terminal body is inserted through the first plug hole, and one end of the terminal body has a contact portion disposed in the slot, the electrical connection module is configured to be plugged into the slot and electrically connected to the contact portion, another end of the terminal body is connected to the limiting structure limited in the auxiliary slot, and the conductive terminal is electrically connected to the circuit board.

2. The lighting effect module of claim 1, wherein, the engaging module comprises a connection block and a plurality of first engaging pieces disposed on two sides of the connection block; the housing is provided with an engaging hole, and a first engaging piece of the plurality of first engaging pieces disposed at one side of the connection block is configured to engage and cooperate

with the engaging hole.

3. The lighting effect module of claim 2, wherein the plurality of first engaging pieces are disposed on opposite sides of the connection block.
4. The lighting effect module of claim 3, wherein the electrical connection module comprises two male connectors disposed on the engaging module and electrically connected to each other, and a male connector of the two male connectors is configured to be plugged in and cooperate with the female connector.
5. The lighting effect module of claim 3, wherein a second engaging piece is disposed on the female connector and is configured to engage and cooperate with the circuit board.
6. The lighting effect module of claim 3, wherein a cross section of the housing is quadrilateral, and each of at least one sidewall of the housing is provided with the female connector.
7. The lighting effect module of claim 2, wherein a shallow groove is disposed on the housing and is configured to limit the connection block.
8. The lighting effect module of claim 7, wherein the electrical connection module comprises two male connectors disposed on the engaging module and electrically connected to each other, and a male connector of the two male connectors is configured to be plugged in and cooperate with the female connector.
9. The lighting effect module of claim 7, wherein a second engaging piece is disposed on the female connector and is configured to engage and cooperate with the circuit board.
10. The lighting effect module of claim 2, wherein the electrical connection module comprises two male connectors disposed on the engaging module and electrically connected to each other, and a male connector of the two male connectors is configured to be plugged in and cooperate with the female connector.
11. The lighting effect module of claim 2, wherein a second engaging piece is disposed on the female connector and is configured to engage and cooperate with the circuit board.
12. The lighting effect module of claim 2, wherein a cross section of the housing is quadrilateral, and each of at least one sidewall of the housing is provided with the female connector.
13. The lighting effect module of claim 1, wherein the electrical connection module comprises two male connectors disposed on the engaging module and electrically connected to each other, and a male connector of the two male connectors is configured to be plugged in and cooperate with the female connector.
14. The lighting effect module of claim 1, wherein the female connector further comprises an engaging holder, the seat body is further formed with a second plug hole, and the electrical connection module is formed with a clamping groove, wherein one end of the engaging holder is configured to be limited in the second plug hole and another end of the engaging holder is configured to engage and cooperate with the clamping groove.
15. The lighting effect module of claim 1, wherein a cross section of the housing is quadrilateral, and each of at least one sidewall of the housing is provided with the female connector.
16. A light, comprising at least one lighting effect module, wherein each of the at least one lighting effect module comprise: a light module, wherein the light module comprises a housing, a light cover, a female connector, a light-emitting body, and a circuit board, the light cover and the female connector are each disposed on the housing, the light-emitting body and the circuit board are each disposed in the housing, and the light-emitting body and the female connector are electrically connected to the circuit board; and a connection assembly, wherein the connection assembly comprises an electrical connection module and an engaging module, the electrical connection module is configured to be plugged into and cooperate with female connectors of two light modules, and the engaging module is configured to engage with housings of the two light modules to enable the two light modules to be electrically connected and engaged through the connection assembly; wherein the female connector comprises: a seat body formed with a slot, a first plug hole, and an auxiliary slot, and the first plug hole separately communicates with the slot and the

auxiliary slot and a conductive terminal comprising a terminal body and a limiting structure, wherein the terminal body is inserted through the first plug hole, and one end of the terminal body has a contact portion disposed in the slot, the electrical connection module is configured to be plugged into the slot and electrically connected to the contact portion, another end of the terminal body is connected to the limiting structure limited in the auxiliary slot, and the conductive terminal is electrically connected to the circuit board.
